SOPHORNNRET PEOU

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RESEARCH & INTERESTS

Creative, motivated in seeking new experiences in electrical system/software Implementation.
Renewable energy system technology, controls and integration circuit design.

EXPERIENCE

COLLINS AEROSPACE — *FSI Engineer-Electrical Engineer (Project LAV. 3.0)* APR 2025 – Present

- System analysis/design including electrical infrastructure, power distribution, and electrical load analysis.
- Develop detailed electrical schematics, wiring diagrams, and I/O documentation for components/test articles.
- Develop detailed test plans, qualification test plans in accordance with compliance and regulation of industry standard accepted by FAA airworthiness as well as mandatory Boeing quality standard requirement.
- Ensure compliance with appropriate regulations including DO-160, DO-178, Boeing D6.
- Managing verification plans, methods, and testing.
- Resolving problems of hardware and components over the product life cycle/maturity.
- Coordinate and collaborate with other engineers such as mechanical, system, software, and test engineers as well as supply chain specialists and project management, to ensure quality with right requirement standard and the flow of the integration process/delivery.
- Technology research for future improvement of project applications as well as process flow within company business unit/customer.

COLLINS AEROSPACE — *Electrical Design/Electronic Engineer (Project LAV. 3.0)* MAR 2023 – Present

- Design and modify electrical/electronic system & ensure electrical system compliance with FAA airworthiness requirements for airborne equipment (DO-documents).
- Schematic capture/review and analysis, PCB board design from start to fabricating and board bring up. V&V and electronic systems testing.
- Solid-state/mechanical switches & relays knowledge, as well as transistors such MOSFET, IGBTs, and other SCRs.
- Create and review QTP/QTR, qualification review based on analysis reports and similarity, support testing process.
- Create/review/maintain ICD of component and system.
- Electrical component research and selection of parts to support electrical system changes in development/modification.

UNIVERSITY of WASHINGTON — *Graduate/Lab Research Design/Experiment* Sep 2024 – June 2026

- Advanced CMOS Fabrication: Nano/Micro scale CMOS fab process include photolithography, oxidation/diffusion, etching, thin film deposition, characterization, overall microfabrication chemical/materials.
- Electrical/Power Electronic Sys. in Renewable NRG: Solar Array/MPPT/Controls, Battery Controls & Manage.
- Embedded & ML: Ind. Research design project/design cycle of embedded microprocessor system, ML/CNN.

UNIVERSITY of WASHINGTON — *Undergraduate/Lab Research Design/Experiment* Sep 2021 – June 2022

- E-Bike Project: building e-bike electrical backbones include power electronic system/control, Schematic/PCB design, control firmware implementation with TI DSP unit, board bring-up, system integration.
- Power Sustainable System Analysis/Dynamic/Protection: AC system analysis, voltage and frequency controls in power systems, principle of economic dispatch and unit commitment.

EDUCATION

<u>Master of Science in Electrical & Computer Engr:</u> – *University of Washington* **Sep 2023 – June 2026**
Advance CMOS Fabrication, Advance Solar Cell/Renewable Energy Power Electronic Systems, Advance Embedded/ML

<u>Bachelor of Science in Electrical Engr:</u> – *University of Washington* **July 2020 – Dec 2022**
Double Conc.: Power Electronics and Drive, Sustainable Power System

<u>Associate in Science:</u> – *Edmonds College* **Sep 2017 – June 2020**
Atmospheric Science, physics, computer science, Certificate of Java Developer (June 2019)

SOFTWARE & DIGITOOLS

Tools: Altium, MATLAB, LTSpice, PLECS, MULTI-SIM, NX, Fusion 360, AutoCAD, Code Composer
Language and Other Tools: Java, Python, C, SQL, C+, C++, HTML, Microsoft Tools/Visual Studio…